

Question:

Calculate the quantity of heat required to (a) convert a liter of H₂O at 30°C to a liter of water at 60°C, and (b) heat a 1 kg block of aluminum from 30° to 60°C. Assume the specific heat of water and aluminum is, re-spectively, 1 cal/g°C and .215 cal/g°C.

Answer Choices:

- A. 28 Kcal, 6.3 Kcal
- B. 40 Kcal, 5 Kcal
- C. 30 Kcal, 8.58 Kcal
- D. 25 Kcal, 8 Kcal
- E. 29 Kcal, 7.77 Kcal
- F. 34 Kcal, 6.9 Kcal
- G. 30 Kcal, 6.45 Kcal
- H. 32 Kcal, 7 Kcal
- I. 35 Kcal, 7.5 Kcal
- J. 31 Kcal, 6.15 Kcal

Reasoning + Answer:

<think> For part (a), we need to calculate the heat required to convert 1 liter of water from 30°C to 60°C. Assuming the density of water is 1 g/mL, 1 liter of water is 1000 g. The heat required is calculated using the formula $Q = m * c * \Delta T$, where m is mass, c is specific heat, and ΔT is the change in temperature. For water, $Q = 1000\text{ g} * 1\text{ cal/g°C} * (60\text{°C} - 30\text{°C}) = 30000\text{ cal}$ or 30 kcal.

For part (b), we need to calculate the heat required to heat 1 kg of aluminum from 30°C to 60°C. Since 1 kg = 1000 g, the heat required is $Q = 1000\text{ g} * 0.215\text{ cal/g°C} * (60\text{°C} - 30\text{°C}) = 2150\text{ cal}$ or 6.3 kcal.

Therefore, the correct values are 30 kcal for water and 6.3 kcal for aluminum.

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